AI 360: Deep Learning
Group Project Proposal

Test-Time Adaptation for Welding Anomaly Detection

Group: Grad Student Descent

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1 Group Composition

Team Member	Student ID
Jaden Majid	94916335
Hieu Le	76067321
Chris Tardy	33171422
Irene Berezin	43769694

2 Introduction and Problem Definition

We address the detection of weld defects from the acoustic signature of the welding arc, using only good welds for training, and keeping it reliable when deployment conditions change.

Large fabrication projects (e.g. shipbuilding) involve thousands of welds, typically checked by post-weld inspection of a sampled subset. Detecting defects during welding would allow immediate rework and more targeted inspection. However, a deployed detector meets unseen stations, microphones and noise, and labelled data from each new site is rarely available. The project is sponsored by Sonibel Instruments who are building commercial solutions to this problem, and from whom we intend to receive proprietary data for test-time adaptation, under an NDA governing its handling and release.

Welding audio is signal rich and defects are diverse and rare. Self-supervised audio encoders can model normal welds without defect labels, and test-time adaptation (TTA) [5, 6, 7, 11] can correct for distribution shift using only unlabelled test data. An academic-use welding dataset with a baseline method is available from Intel [1], and competitions such as DCASE [2] publish state-of-the-art methods, including the one proposed for this project [3].

Our research question is whether the DCASE 2026 winning method (MERL [3]) outperforms the Intel autoencoder baseline for acoustic weld-defect detection. Furthermore, can TTA recover its performance on shifted welding audio without labels?

The core deliverable is a MERL-style detector that exceeds the published audio-only AUC of 0.846 on the Intel welding benchmark [1]. Extensions are (i) TTA on constructed shifts and Sonibel data, (ii) real-time (online) detection, and (iii) audio–video fusion.

3 Related Work and Background

Proof of Concept and Baseline. The Intel welding benchmark [1] is a dataset of both the audio and video streams of welding jobs, classified as correct or as one of 11 real anomalies. Its audio

model is a 1D CNN autoencoder on FFT frames of 192 kHz audio. Test AUC (how well anomaly scores rank defective welds above good ones) is 0.846 for audio, 0.898 for video, and 0.918 fused. Audio is weakest on burnthrough (0.70), warping (0.76), excessive penetration and spatter (both 0.78). It should be noted that the published Intel dataset only supports audio up to 16 kHz, so we would re-implement Intel's method on their public data to determine fair baselines.

The Gap. DCASE Task 2 [2] is an annual challenge on detecting faulty machines from sound with only normal training data. Recent top systems use pretrained audio encoders (BEATs, EAT, CLAP), fine-tune them on auxiliary labels, and score by k-NN distance to normal embeddings. These consistently beat autoencoder baselines but have not been tested on welding. The 2026 winner, MERL [3], scored 66.2% against 57.7% for the autoencoder baseline. Our objective is to take the Intel dataset and beat its baseline with MERL's method.

4 Datasets

We use the Intel Robotic Welding Multimodal Dataset [1]: 4,040 robotic arc welds from an automotive production floor, with audio, video, sensor time series and post-weld images. We focus on audio. Defects span 11 categories, including porosity, burnthrough, spatter and lack of fusion. We reuse Intel's splits so results are directly comparable (Table 1); training uses good welds only. Audio is resampled to each encoder's input rate and converted to its expected log-mel input; the baseline uses FFT frames, following [1]. Access has been approved and we have the dataset; labels denote the intended defect, so some label noise is expected. Sonibel's proprietary data for TTA will be handled under an NDA. The Intel dataset is the only public welding dataset with audio and a published anomaly detection baseline; Sonibel's data provides a realistic shift for TTA.

Split	Good welds	Defective welds
Train	576	0
Validation	122	1,610
Test	121	1,611

Table 1: Intel dataset splits.

5 Available Hardware and Computational Resources

Chris is a graduate student in the PLAI group at UBC and has access to its GPU cluster. The cluster includes several RTX 2080 Ti and A5000 GPUs for model experiments and training. Jaden has a home PC with an RTX 5070 Ti that he can donate for runs that would exceed Colab runtime limits. Since we start from frozen pretrained encoders with light fine-tuning, these resources are sufficient.

6 Proposed Model and Methodology

Model and Starting Point. We use three transformer audio encoders (BEATs, EAT and EAT-large), self-supervised on AudioSet, chosen because public checkpoints are available and the leading DCASE systems use them. As a baseline, we re-implement Intel's 1D CNN autoencoder [1]; its code is not public, but the paper lists the full architecture and training settings. All three encoders

are used pretrained, first frozen. We then fine-tune them on good welds with an auxiliary task: classifying each weld's process settings (weld type, current, voltage). Only the Intel autoencoder is trained from scratch, on the released 16 kHz audio (the paper used 192 kHz), to obtain a new, directly comparable baseline. We pair the encoders with MERL's scoring method [3] and average anomaly scores across encoders.

Proposed Approach. Inputs are mono 16 kHz weld recordings, converted to the log-mel spectrogram each encoder expects, and to FFT frames for the Intel baseline. The method has five steps:

1. We reproduce the Intel baseline at 16 kHz, scoring each weld recording by its reconstruction error and measuring test AUC.
2. Each frozen encoder maps a weld to embeddings; its anomaly score is the distance to its nearest neighbours among the 576 good training welds, so a weld unlike any good weld scores high. We measure AUC for an ensemble of the 3 encoders.
3. We fine-tune the encoders on the auxiliary settings-classification task, so their embeddings separate welding-relevant differences. The classifier head is then discarded.
4. We repeat k-NN detection with MERL's scoring. Rather than comparing whole welds, we compare each frequency band separately and average, so an unfamiliar mix of normal sounds is not flagged; we emphasise moments that differ most from the rest of the weld, so a brief defect is not averaged away; and we adjust scores for how tightly good welds cluster. We measure AUC for an ensemble of 3 encoders $\times$ 3 fine-tuning seeds, with hyperparameter search & CV if necessary.
5. As an extension, we apply test-time adaptation (TTA) to adapt the encoders to Sonibel data, a distribution shift from the Intel recordings. If Sonibel data is unavailable, we will instead construct corrupted test sets in the spirit of ImageNet-C and CIFAR-10-C [10], using common audio corruptions (e.g. background noise, reverberation, gain changes, clipping). TTA methods will be chosen after a literature review; candidates include EATA [8], FOA [9], SAR [7], ActMAD [6], NOTE [11] and BN adaptation [5].

The Intel autoencoder baseline minimises reconstruction error; fine-tuning uses a cosine-similarity classification loss on the auxiliary labels, as in MERL. The k-NN scorer requires no training. Fine-tuning uses low-rank adapters (LoRA) with AdamW; key hyperparameters are k, the pooling weight and the number of class centres. Each weld receives an anomaly score, averaged across the ensemble; online, an alarm is raised when the running score exceeds a threshold set on good validation welds.

Baselines and Evaluation. Baselines are the reproduced Intel 1D CNN autoencoder (reported 0.846 audio AUC) and the frozen pretrained encoders with MERL-style k-NN scoring, which isolates the gain from fine-tuning. Following [1], we report clip-level ROC-AUC overall and per defect category; for online detection, time-to-detection and false-alarm rate. Ablations isolate the encoder, fine-tuning and each scoring component. For TTA, we compare AUC before and after adaptation (BN adaptation [5], ActMAD [6], SAR [7]) on constructed shifts (e.g. added noise, held-out weld types) and Sonibel data. Success is exceeding 0.846 audio-only AUC; a strong result would approach the 0.918 audio–video figure with audio alone.

7 Timeline and Deliverables

Time Period	Milestone / Activities	Expected Deliverable
Nov 6–11	Reproduce Intel baseline	Reproduced baseline AUC
Nov 12–18	Frozen encoders + k-NN (MERL scoring)	Frozen-encoder AUC
Nov 19–25	Fine-tune encoders; hyperparameter sweeps	Fine-tuned AUC beats baseline
Nov 26–Dec 2	TTA on shifted data	AUC before and after TTA
Dec 3–7	Final analysis and report	Final report and slides

Checkpoint (Nov 25): compare the reproduced Intel baseline, frozen and fine-tuned encoders on Intel's splits; revise scope if the baseline cannot be exceeded.

8 Division of Work

We intend to work in parallel where each team member has a primary responsibility, but we contribute across tasks. We intend to merge the best parts of each team member's contributions at the end of each milestone.

Team Member	Primary Responsibilities	Expected Contributions
Jaden Majid	Fine-tuning Encoders	Fine-tuned encoder checkpoints beating frozen on validation AUC
Hieu Le	Reproduce Baseline	Intel method implementation and new AUC baseline
Chris Tardy	Literature Review + TTA	Novel TTA methods, implementation and results vs. baseline
Irene Berezin	KNN implementation & Evals	Frozen-encoder results, k-NN implementation, hyperparameter sweeps

References

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